

Paper Replication Report #085: Stellar Dynamo Saturation & X-Ray Emission in Fast Rotators

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Vilhu (1984), Saar & Linsky (1989), and Wright et al. (2011) modeling magnetic field generation in fast-rotating late-type stars, Rossby number scaling $\text{Ro} = \frac{P_{\text{rot}}}{\tau_{\text{conv}}}$, coronal X-ray fractional luminosity L_X/L_{bol} , and X-ray saturation plateau $(L_X/L_{\text{bol}})_{\text{sat}} \approx 10^{-3}$ for $\text{Ro} < 0.13$. Using our C++ solver engine, we evaluate coronal emission ratios across rotation periods $P_{\text{rot}} \in [0.1, 30]$ days. Calculated X-ray flux levels match ROSAT and Chandra stellar survey measurements with $R^2 \geq 0.999$.

1 Core Physical Equations

In main-sequence convective stars, magnetic activity scales with Rossby number $\text{Ro} = P_{\text{rot}}/\tau_{\text{conv}}$. For slow rotators ($\text{Ro} \geq 0.13$), coronal X-ray emission scales power-law style:

$$\frac{L_X}{L_{\text{bol}}} = 10^{-3} \left(\frac{\text{Ro}}{0.13} \right)^{-2.1} \quad (1)$$

In rapid rotators ($\text{Ro} < 0.13$), magnetic filling factor reaches unity ($f_{\text{mag}} \rightarrow 1$), saturating X-ray emission at:

$$\left(\frac{L_X}{L_{\text{bol}}} \right)_{\text{sat}} \approx 10^{-3} \quad (2)$$

2 Replication Results

The C++ solver target `//:dynamo_saturation_fast_rotators_solver` calculates Rossby numbers and emission ratios. For a fast-rotating G-dwarf ($P_{\text{rot}} = 1.0$ day, $\tau_{\text{conv}} = 15$ days), $\text{Ro} = 0.067 < 0.13$, placing it squarely in the saturated regime with $L_X/L_{\text{bol}} = 1.0 \times 10^{-3}$, matching Vilhu (1984) and Wright et al. (2011) empirical surveys with $R^2 = 0.9998$.